THE LIMIT TOPOLOGY IS I-ADIC, IN THE SHARPENED FORM USING THE STABLE IMAGES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's own argument: it states in the first sentence that it runs like the companion statement of this chapter, and then carries out the paragraph in which the sharpened hypothesis is used; that first sentence is retained verbatim inside the counted proof span and is not expanded by this card.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- I-adic and limit topologies on the limit of an inverse system of modules. Same chapter, the comparison of the I-adic topology with the limit topology on M = lim H^p(C, F_n). Supplies the two topologies compared in the statement and the neighbourhoods of the limit topology.
- Graded module structure and ascending chain condition on the stable images. Same chapter, the graded module structure of the direct sum of the stable images N_n over the ring of the powers of I, and the ascending chain condition hypothesis. Supplies the finite generation used in the proof, which is the sharpening with respect to the companion statement.

Source reference. Original lemma 0GYS: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4865–4865. Navigation pointer invoked at proof line 4965; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0GYR: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4789–4789. Navigation pointer invoked at proof

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line 4972; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let I be an ideal of a ring A. Let $\mathcal{C}$ be a site. Let*

$$\ldots \to \mathcal{F}_3 \to \mathcal{F}_2 \to \mathcal{F}_1$$

be an inverse system of sheaves of A-modules on $\mathcal{C}$ such that $\mathcal{F}_n = \mathcal{F}_{n+1}/I^n\mathcal{F}_{n+1}$. Let $p \geq 0$. Given n define

$$N_n = \bigcap\nolimits_{m \geq n} \mathrm{Im}\left(H^p(\mathcal{C}, I^n\mathcal{F}_{m+1}) \to H^p(\mathcal{C}, I^n\mathcal{F}_{n+1})\right)$$

If $\bigoplus N_n$ satisfies the ascending chain condition as a graded $\bigoplus_{n\geq 0} I^n/I^{n+1}$-module, then the limit topology on $M = \lim H^p(\mathcal{C}, \mathcal{F}_n)$ is the I-adic topology.

Proof. The proof is exactly the same as the proof of Lemma 0GYS. In fact, the result will follow from the arguments given there as soon as we show that $\bigoplus N_n$ is a graded $\bigoplus_{n\geq 0} I^n/I^{n+1}$-submodule of $\bigoplus H^{p+1}(\mathcal{C}, I^n\mathcal{F}_{n+1})$ and that $F^n/F^{n+1} \subset H^p(\mathcal{C}, \mathcal{F}_{n+1})$ is contained in the image of $N_n \to H^p(\mathcal{C}, \mathcal{F}_{n+1})$. In the proof of Lemma 0GYR we have seen the statement on the module structure.

Let $t \in F^n$. Choose an element $s \in H^p(\mathcal{C}, I^n\mathcal{F}_{n+1})$ which maps to the image of t in $H^p(\mathcal{C}, \mathcal{F}_{n+1})$. We have to show that s is in N_n. Now F^n is the kernel of the map from $M \to H^p(\mathcal{C}, \mathcal{F}_n)$ hence for all $m \geq n$ we can map t to an element $t_m \in H^p(\mathcal{C}, \mathcal{F}_{m+1})$ which maps to zero in $H^p(\mathcal{C}, \mathcal{F}_n)$. Consider the cohomology sequence

$$H^{p-1}(\mathcal{C}, \mathcal{F}_n) \to H^p(\mathcal{C}, I^n\mathcal{F}_{m+1}) \to H^p(\mathcal{C}, \mathcal{F}_{m+1}) \to H^p(\mathcal{C}, \mathcal{F}_n)$$

coming from the short exact sequence $0 \to I^n\mathcal{F}_{m+1} \to \mathcal{F}_{m+1} \to \mathcal{F}_n \to 0$. We can choose $s_m \in H^p(\mathcal{C}, I^n\mathcal{F}_{m+1})$ mapping to t_m. Comparing the sequence above with the one for $m = n$ we see that s_m maps to s up to an element in the image of $H^{p-1}(\mathcal{C}, \mathcal{F}_n) \to H^p(\mathcal{C}, I^n\mathcal{F}_{n+1})$. However, this map factors through the map $H^p(\mathcal{C}, I^n\mathcal{F}_{m+1}) \to H^p(\mathcal{C}, I^n\mathcal{F}_{n+1})$ and we see that s is in the image as desired. □